Molecular Assessment and Validation of the Selected Enterococcal Strains as Probiotics

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RESEARCH

Molecular Assessment and Validation of the Selected Enterococcal Strains as Probiotics

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Abstract
Probiotics are live microorganisms which confer health benefits to the host. Lactic acid bacteria (LAB) are used as probiotics since decades. Enterococci being the member of LAB have proven probiotic strains; therefore, this study was aimed at finding out the potential probiotic candidates from the pool of locally isolated strains. For initial screening, one hundred and twenty-two strains were selected and subjected to different confirmatory and phenotypic tests to choose the best strains that have potential probiotic criteria, i.e., no potential virulence traits, antibiotic resistance, and having tolerance properties. Keeping this criterion, only eleven strains ($n=11$) were selected for further assessment. All virulence traits such as production of hemolysin, gelatinase, biofilm, and DNase were performed and not found in the tested strains. The molecular assessment indicates the presence of few virulence-associated genes in *Enterococcus faecalis* strains with variable frequency. The phenotypic and genotypic assessments of antibiotic resistance profile indicate that the selected strain was susceptible to ten commonly used antibiotics, and there were no transferrable antibiotic resistance genes. The presence of CRISPR-Cas genes also confirmed the absence of antibiotic resistance genes. Various enterocin-producing genes like *EntP*, *EntB*, *EntA*, and *EntQ* were also identified in the selected strains which make them promising probiotic lead strains. Different tolerance assays like acid, NaCl, and gastric juice tolerance that mimic host conditions was also evaluated by providing artificial conditions. Cellular adhesion and aggregation properties like auto- and co-aggregation were also checked and their results reflect all in the favor of lead probiotic strains.

Keywords *Enterococcus* · Probiotics · CRISPR-Cas · Antibiotic resistance · Enterocins

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Introduction

Genus *Enterococcus* is the third largest genus in the class Lactic acid bacteria (LAB) that produces antimicrobial proteins and peptides, i.e., bacteriocins [1]. They have a ubiquitous distribution and found in shallow and deep water, soil, air, plants, sewage, vegetable, foods, clinical environments, and gut of human and other animals [2, 3]. Characteristically, this genus is Gram positive, catalase negative [2], aero-tolerant, and fermentative bacteria [4] that can tolerate and survive in harsh environmental conditions, like high pH, salt, and temperature and also show resistance to desiccation [5, 6]. The presence of genomic plasticity makes them a very useful microorganism and helps them to use in various biotechnological and food-associated processes. They also impart organoleptic properties to foods [4, 7].

Initially, enterococci were considered common commensal microbes of the gut but with the passage of time, slight pathogenicity was developed and currently, they become the

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2nd most common pathogens of hospital-associated infections (HAI) [2, 8]. Both intrinsic and acquired antibiotic resistances were identified which make them a causative agent of multiple disease like urinary tract, pelvic, belly, burn, and surgical wound infections and infections related to medical instruments [9, 10]. The virulent profile of enterococci is not so high but various factors are identified which increase their pathogenicity [9]. Predominant among them are cytolysin (hemolysin), gelatinase, serine protease, hyaluronidase, aggregation substances, adhesion to collagen surface proteins, and biofilm formation substances [5, 10]. Being the member of LAB, enterococci can produce different types of small and low molecular weight proteins and peptides called enterocins. These substances are cationic and hydrophobic in nature and show activity against various pathogenic microorganisms [9, 11].

Probiotics was first described by Lilley and Stillwell [12], and it is defined as “those mono or mixed live microorganism which when consumed in such amount that confers health of the host” [9, 13, 14]. According to the definition of Food and Agriculture Organization (FAO), World Health Organization (WHO), and International Scientific Association for probiotics and Prebiotics (ISAPP), probiotics are “live microorganisms which when administered in adequate amounts confer a health benefit on the host” [12, 15, 16]. Not all microorganisms are used as probiotics; hence, the WHO, FAO, and European Food Safety Authority (EFSA) proposed a selection criteria, such as the production of antimicrobial substances, attachment to human intestinal cells, immunity altering ability [17], and viability at gastrointestinal tract (GIT) conditions, i.e., low pH, pepsin, pancreatic, bile salts [12, 15], human origin, unable to translocate in the intestinal epithelium [14], susceptible to antibiotics, kill other pathogen, and enterocin production [9]. Probiotics have a number of health benefits in human and animals, like they strengthen the immune system, improving intestinal barrier function, reducing hypersensitivity [18], help in antibiotic-associated diarrhea [19], reduced metabolic disorders, altering pain perception, improve food digestion, used in food additive, treat T-cell imbalance [20], also help in the treatment of necrotizing enterocolitis, reduced cholesterol, and help in cancer prevention [18, 21, 22].

Widely used probiotics belongs to the genus *Lactobacillus* and *Bifidobacterium*, while enterococci are used occasionally [17, 23]. The limited usage of enterococcal species is due to the presence of different virulence traits and low- to high-level antibiotic resistance which make them an ambiguous entity for probiotic potential [9]. Although enterococcal species are currently used in several industries like, pharma, leather, and food [4, 24, 25], which are mostly attributed by the production of antimicrobial proteins and peptides (namely, enterocins), *Enterococcus faecium* is widely studied for their probiotic potential because of their unique features and ability of multiple enterocins productions [13]. Currently available enterococcal probiotics are used for the treatment and prevention of diseases like irritable bowel syndrome (IBS), antibiotic-associated diarrhea (AAD), and having antitumor, cholesterol-lowering, and immune modulatory effects. These also show activity against aquatic pathogens, improved fish immunity, and having antiviral and antimicrobial activities [26]. Generally, enterococcal species required more care, proper identification, and high authenticity for their probiotic potential.

By covering multidimensional aspects of *Enterococcus*, we have collected and established an enterococcal biobank ($n=2600$) isolated and characterized from the local environments [5, 10, 27]. A huge data set is available describing their prevalence, virulence profile, and antibiotic resistance, but very little is known about their probiotic potential. Thus, in this study, selected enterococcal strains were tested for virulence traits, antibiotic resistance, and in vitro assessment to establish their probiotic potential.

Material and Methods

Bacterial Strain Selection

Bacterial strains were selected from our established enterococcal species biobank originated from different environmental niches like soil, poultry, dairy, fresh water, and sewage and from clinical sources from different areas of Karachi, Pakistan [3, 10]. The preliminary selection from this huge data set ($n=2600$) was based on non-clinical source, enterocin production, free from apparent virulence traits, and antibiotic resistance [28]. Based on these aforementioned criteria ($n=122$), strains were selected for their initial probiotic potential. Market available enterococcal probiotic *E. faecium* NF strain (RG Pharmaceutica Pvt. Ltd., Karachi) was used as positive control. All the selected strains were preserved in brain heart infusion (BHI) broth with 20% glycerol. From preserved isolates, working culture was prepared for further characterization as described earlier [27].

Genomic DNA Preparation and Identification of the Strains

Total genomic DNA of the selected strains were extracted by CTAB method [29], for use as template in genome typing or all single and/or multiplex-PCR analysis. Primer was reconstituted according to their molar concentration using Tris–EDTA buffer, pH 8.2, as described earlier by us [3, 27]. Selected strains were reconfirmed and identified by multiplex-PCR using 16S rRNA (genus specific) and *SodA* (species specific) gene markers as described earlier [27, 30].

A known set of ATCC strains was used as positive controls. Phenotypically, the strains were reconfirmed by culturing on Slanetz and Bartley agar (SBA) media and by various tests like catalase and urease, according to the method described earlier by us [3, 10].

RAPD-PCR Analysis

Random amplified polymorphic DNA (RAPD) PCR was used to determine the genetic relatedness and heterogeneity followed by dendrogram construction analysis using M13-specific oligonucleotide primer as recently described by Rehman et al. [3].

Phenotypic and Genotypic Assessments of Virulence Traits

According to WHO and FAO guideline for probiotic selection, the strains must be free from any virulence trait–related genes [31]. Phenotypically, the virulence traits like biofilm formation, gelatinase, catalase, DNAse, and hemolysis were checked by the method described by Hasan et al. [10]. Similarly, genes responsible for these traits, i.e., *esp*, *agg*, *ace*, *acm*, *EfaAfs*, *EfaAfm*, *cylLl*, *cyl A*, *cyl B*, *cyl M*, *gelE*, *sprE*, *kat*, *lip fs/fm*, *eep*, *ccf*, *cob*, *cdf*, and *hyl* [32], were also screened as described earlier [3, 27].

Evaluation of Enterocin Production and CRISPR-Cas-Associated Genes

Phenotypically, bacteriocinogenic activity was evaluated by the established stab overlay method [3, 7]. Briefly, overnight grown test strain was evaluated against sensitive or indicator/test strain (e.g., *Listeria monocytogenes* (ATCC13932) and *E. faecium* N283) and the activity was measured as zone of inhibition in millimeter [27]. The molecular assessment of known enterocin genes was investigated by using their specific primers as previously described by us [3]. *EntA*, *EntB*, *EntP*, *EntQ*, *Ent ef1097*, *EntL*, Ent L50, and Ent SE-K4 were screened [33, 34]. The PCR product was confirmed by DNA agarose gel electrophoresis and visualized by an UV transilluminator (UVP, UK). DNA from well-established enterocin-producing strains were used as positive control.

The presence of CRISPR-Cas genes in the selected strains was investigated according to Lindenstrauß et al. [35] with little modification. Briefly, for each experiment, 12 µL of total reaction volume was prepared with 0.5 µL of primer (16 µM) and 2 µL of genomic DNA. CRISPR-Cas genes CRISPR1, CRISPR1-cas csn1, CRISPR2, CRISPR3, and CRISPR3-cas csn1 were investigated.

Assessment of Antibiotic Resistance Profile

The evaluation of antibiotic resistance genes is the essential selection criteria for probiotic strains [34]. Antibiotic concentrations were selected in the light of the guideline proposed by Clinical and Laboratory Standard Institute (CLSI) [36], EFSA, and other authors like [31, 32, 37]. Both phenotypic and genotypic antibiotic resistances were determined by the methods recently described by us [10, 30]. The selected antibiotics include tetracycline, erythromycin, linezolid, vancomycin, chloramphenicol, gentamycin, penicillin, ampicillin, teicoplanin, and ciprofloxacin. The results were shown as susceptible (S), resistant (R), and intermediate (I) resistant.

ATR-FTIR and SDS-PAGE Analysis

For attenuated total reflectance-Fourier transform infrared (ATR-FTIR) analysis, the samples were prepared as described earlier [38] with slight modification. Briefly, bacterial culture was grown in BHI broth for 24 h without shaking. After centrifugation (12,000 rpm for 15 min), the cell pellets were washed three times with DI water and lyophilized. FTIR instrument (TENSOE II, Bruker, Germany) with ATR single-reflection diamond head was used to record FTIR spectra by measuring 32 scans at 4 cm^{-1} resolutions in the range of 4000 to 500 cm^{-1}. Similar experimental conditions were applied for every sample. The operating software OPUS version 7.5 (Bruker Ettlingen, Germany) was used for data analysis.

For protein finger printing, samples of selected strains were subjected for SDS-PAGE analysis under reducing conditions using 12% polyacrylamide gel [39]. Briefly, overnight BHI grown culture was centrifuged (at 12,000 rpm for 15 min) and the supernatants were discarded, while the pellets were washed thrice with DI water. Finally, pellets were suspended directly in reducing sample diluting buffer, incubated at 100 °C for 10 min, and then centrifuged for 15 min at 14,000 rpm. Obtained cell-free supernatants were subjected for SDS-PAGE analysis. Molecular weight marker (Unstained Protein Molecular Weight Marker, Fermentas, USA) was also loaded as size standard. The gel was stained overnight in colloidal Coomassie brilliant blue G-250 and then distained with DI water.

Gastric Juice, Acid, and Salt Tolerance

Tolerance of the selected strains to gastric juice was assessed by the method of Fu et al. [40]. Overnight, test strains were centrifuged with 10,000 rpm for 5 min and washed three times with phosphate-buffered saline (PBS, pH 7.2). Cell-free supernatant (CFS) was filter via 0.22-µm filter. Aliquots of gastric juice having different pH (pH1.0, pH2.0, pH3.0, and pH4.0) were prepared and supplemented with supernatant of test strains and then added into a 96-well sterile plate. The absorbance (OD_{600}) was measured for 4 h with 1-h interval. Gastric juice was prepared by dissolving NaCl 2 g/L and pepsin 3 g/L, while adjusted the pH with 1 N HCl accordingly.

The acid tolerance of the selected strains was established by the modified method of Pieniz et al. [41]. Briefly, overnight grown culture was standardized (OD_{600}) to 1.0 ± 0.05. The pH was adjusted to different values (pH2.0, pH4.0, pH7.0, pH9.0, pH12.0) with 1 N HCl/NaOH and then added to a sterile 96-well plate. This pH-modified BHI was inoculated with freshly grown culture and their absorbance (OD_{600}) was measured at different time intervals for 24 h. pH 7.0 was used as control.

For NaCl tolerance, the selected strains were treated against various salt concentrations as described recently [42]. BHI broth supplemented with different concentrations of NaCl (1%, 3%, 5%, and 7%) was inoculated with fresh culture in a 96-well plate, and their absorbance (OD_{600}) was measured at different time intervals for 24 h. BHI broth without NaCl was used as control.

Antimicrobial Activity of the Selected Strains

Antimicrobial potential of the selected strains was performed as described by Rehman et al. [3]. Three microliters of the test strains was spotted on freshly grown indicator strain on BHI agar and incubated at 37 °C for 24 h. Formation of a clear zone of growth inhibition around the test strain was considered positive and measured in millimeter. The indicator strain against which the selected strains were screened, include both Gram-positive like *Micrococcus luteus* (ATCC10242), *Lactobacillus* sp. (milk), *Staphylococcus aureus* (ATCC25923), *E. faecalis* (ATCC51299), *Enterococcus hirae* (ATCC8043) (Food), *Enterococcus casseliflavus* (Food), *E. faecalis* (ATCC29212), *E. faecium* (ATCC6569), *Enterococcus durans* (ATCC6056), *Sensitive* (*E. faecium* NA283), and *Listeria monocytogenes* (ATCC13932) and Gram-negative bacteria like *Pseudomonas aeruginosa* (ATCC27853), *Klebsiella pneumonia* (ATCC700603), *Enterobacter aerogenes* (ATCC13048), *E. coli* (ATCC13047 and ATCC32518), and *Proteus mirabilis* (UTI).

Auto- and Co-aggregation

Auto- and co-aggregation ability of the selected strains was established according to the method described earlier [6, 37] respectively. In both assays, the overnight test culture was centrifuged at 4000 rpm for 20 min, the supernatant was discarded, and the pellets were washed and suspended in PBS. Initial absorbance ($OD0_{600}$) was record for both assays. For auto-aggregation, the samples were incubated for 1 h and again centrifuged at 3000 rpm for 20 min and their absorbance was recorded, while for co-aggregation, suspension of both probiotics and pathogenic strains was taken in the same volume and incubated at 37 °C without shaking; then, their final absorbance was measured at 24 h. The percent auto- and co-aggregations were calculated as follows:

$$\%\text{age co-aggregation} = \frac{(\text{OD0} - \text{OD1})}{\text{OD0}} \times 100$$

where OD0 refers to initial value, and OD1 refers to reading after incubation.

Cell Surface Hydrophobicity

Cell surface hydrophobicity was determined by the adherence of selected strains to hydrocarbon as described by Bhardwaj et al. [43]. Overnight culture was centrifuged at 10,000 rpm for 10 min. The pellets were washed and suspended in phosphate urea magnesium sulfate (PUMS) buffer, and their initial absorbance ($OD0_{450}$) was recorded. Hydrocarbon (xylene) was mixed in the ratio of 3:1 and pre-incubated at 30 °C for 10 min, followed by vortexing and 1-h incubation at room temperature. The aqueous phase was taken and their absorbance was measured. The percent hydrophobicity was calculated by the same aforementioned formula.

Cytotoxicity Determination of Bacteria

Cytotoxicity assays was performed according to the method recently reported by Dowdell et al. [44] with slight modification. Briefly, human oral squamous cell carcinoma (CAL-27; ATCC, USA) cells were seeded in a 96-well plate (12×10^3/well) and incubated at 37 °C for 24 h in the presence of 5% CO_2. Next day, the test strains were centrifuged at 10,000 rpm for 15 min in order to collect cell-free supernatants. The supernatant was added into each well, and the plates were incubated at 37 °C for 48 h, while wells without test sample was used as a control. After incubation, the supernatant was removed, well was washed, and media were added containing 0.5 mg/mL of 3-(4, 5-di-methyl thiasol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) solution (Merck, Kenilworth, NJ, USA) to each well and incubated for 3 h in the dark. The MTT was removed carefully and 100 µl di-methyl sulfoxide (DMSO; Merck, Kenilworth, NJ, USA) was added to solubilize the crystals, and their absorbance was measured at 570 nm. The percentage cell viability was calculated by the following formula:

$$\% \text{ cell viability} = \left[(\text{OD sample} \times 100) / \text{OD control} \right]$$

Adherence Assay

Adherence assay was performed using human colorectal adenocarcinoma cell lines (HT-29; ATCC, USA). The bacterial adhesion was determined as previously described [45] with some modification. Briefly, 7×10^4 cells were seeded in fresh DMEM to prepare monolayer of HT-29 cells on eight well-chambered slide (Lab-Tek, USA). Bacterial inoculum (5×10^8 cells/mL) FITC-conjugated and non-conjugated bacterial samples were prepared and inoculated on chambered slide containing HT-29 monolayer. Chambered slides were prepared with following combination per well containing HT-29 cells. The 1st and 2nd wells were inoculated with non-FITC-labeled bacteria, 3rd and 4th wells contain FITC-labeled bacteria, 5th and 6th wells contain only eukaryotic cell, and 7th and 8th well served as bacteria and medium controls, respectively. After inoculations, slides were incubated for 4 h at 37 °C in a humidified atmosphere containing 5% CO_2 (Heal Force, Korea).

The monolayers were washed thrice with sterile PBS then fixed with absolute methanol (Tedia, USA) for 3–5 min. After fixation, the slide was mounted with Mowiol®4–88 (Carl Roth, Germany) mounting media. Slide was visualized microscopically using a Nikon Ti-2 microscope (Nikon, Japan) on DIC and florescence mode. Images were processed via NIS-Elements Advanced Research software (ver. 3.22.09). Eight to 10 fields were examined for adherence and reconfirmed by florescence microscopy with FITC-labeled bacteria.

Statistical Analysis

All the experiment was performed in triplicate and the results were presented as mean ± SD. Statistical significance of the data was analyzed using one-way analysis of variance (ANOVA) on IBM SPSS statistics V21.0. The data were considered significance with value $p < 0.05$.

Results

Identification of Selected Strains

Identification of the subjected strains ($n = 122$) as performed by both phenotypic and genotypic methods revealed the highest percentage from food source (52%) followed by milk (18%) and water/sewage (13%). Species prevalence indicates that *E. faecium* (60%) and *E. faecalis* (36%) were the most distributed species. Besides, these two strains *E. hirae* and *E. casseliflavus* were also isolated (Fig. S1). Based on apparent desirable probiotic characteristics, only eleven strains ($n = 11$) were subjected for further in-depth evaluation.

Genetic Heterogeneity by RAPD-PCR, ATR-FTIR, and SDS-PAGE Analysis

RAPD-PCR analysis followed by dendrogram construction shows that the selected strains were divided into two clades (A and B). Clade-A is further divided into two groups A1 and A2. Group A1 is comprised of both *E. faecalis* and *E. faecium* strains, while group-A2 has only *E. faecium* strains. In Clade-B, the group b1 has both *E. faecalis* and *E. faecium* strains, while b2 has only *E. faecium* strains. The overall similarity index is ranging from 82 to 96%, hence indicating their close relationships (Fig. 1A, B). In order to complement, protein finger printing was also performed by subjecting the whole-cell protein extracts to SDS-PAGE analysis. Obtained results show a consistent and identical band pattern in the range of 40 to 50 kDa, followed by a variable region between 10 and 20 kDa in all selected strains with reference to control strains (*E. faecalis* ATCC 51299 and *E. faecium* ATCC 6569). Interestingly, a ~ 15 kDa band was uniquely observed in the tested strains but not detected in both control ATCC strains (Fig. S2).

Likewise, FTIR spectra were also recorded in the range of 4000 to 500 cm^{-1} as it was previously reported that specific signals for probiotics are located at 2845 and 1929 cm^{-1} [46]. In our results, probiotic-specific signal was observed at 2924 cm^{-1}. Regions above 3200 cm^{-1} show the presence of fatty acid contents, while peaks at 1700 cm^{-1} indicate typical amide band of proteins. Peaks in the range of 1300 to 900 cm^{-1} are known as "fingerprint" region of the lactic acid bacteria. All the data was analyzed with reference to control strain *E. faecium* NF, a market-available probiotic strain (Fig. 2A, B).

Phenotypic and Genotypic Assessment of Virulence Traits

Phenotypic analysis of the selected strains shows that all *E. faecium* strains were free from virulence traits compared to *E. faecalis* strains. Gelatinase, lipase, catalase, DNAse, and more importantly hemolysin production was not detected in any strain (Table S1). The absence of hemolytic activity was also complemented by the absence of cytotoxicity tested against CAL-27 cell line using standard MTT assay. Molecular assessment of virulence-associated genes like cytolysin (*cylLl*, *cyl A*, *cyl B*, *cylM*, *cylLS*), gelatinase (*gelE*), serine protease (*sprE*), insertion sequence (*IS16*), enterococcal surface proteins (*esp*), aggregation substances (*agg,asa1*), endocarditis-associated (*EfaAfs*, *EfaAfm*), collagen binding (*ace*, *acm*), hyaluronidase (*hyl*), sex pheromone (*eep*,

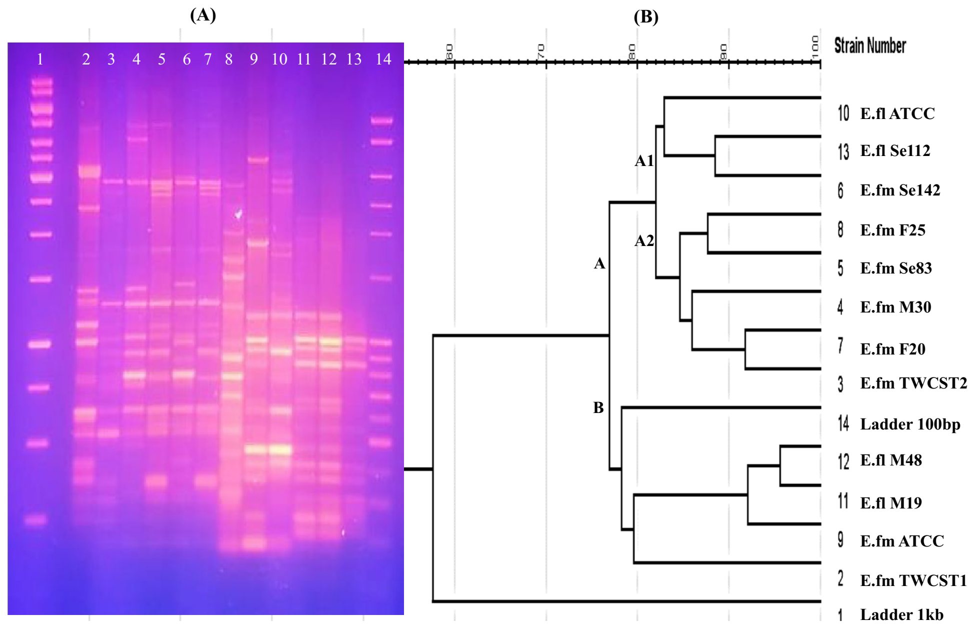


Fig. 1 Unweighted pair group method using arithmetic average (UPGMA)–based dendrogram derived from RAPD-PCR patterns generated with M13 primer of the selected enterococcal strains isolated from different environments

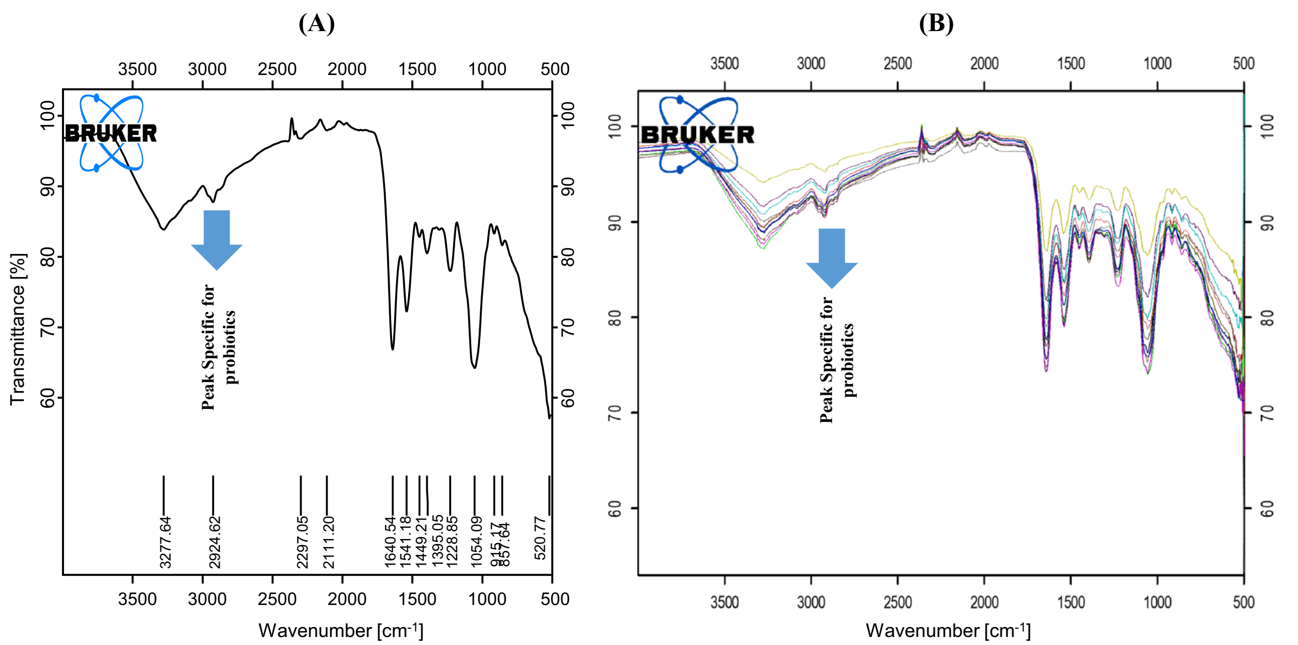


Fig. 2 ATR-FTIR spectra of the selected enterococcal strains recorded at 4 cm^{-1} resolutions of 32 scans in the range of 4000 to 500 cm^{-1}. **A** Control probiotic *E.fm* NF and **B** other tested enterococcal strains

ccf, *cob*, *cdf*) catalase (*kat*), and lipase-associated (*lip fs/fm*) genes were screened for strain safety, which was not observed in most *E. faecium* strains, while *E. faecalis* harbor few virulence genes. Biogenic amine-associated genes (i.e., *hdc*, *tdc*, *odc*) were also absent in all tested strains. Representative gels of major virulence traits are shown in Fig. S3, while their complete molecular assessment is summarized in Table 1.

Phenotypic and Genotypic Assessment of Antibiotic Resistance

The European Food Safety Authority (EFSA, 2018) antibiotic guidelines for probiotics recommend that cell wall synthesis inhibitors, like ampicillin and vancomycin, and protein synthesis inhibitors, like erythromycin, gentamycin, chloramphenicol, and tetracycline, should be screened for

Table 1 Molecular assessment of virulence traits, sex pheromones, adhesion substances, and biogenic amine genes of the selected enterococcal strains

Strains											
Genes	*E.fm* NF[a]	*E.fm* TWCST1	*E.fm* TWCST2	*E.fm* M30	*E.fm* Se83	*E.fm* Se142	*E.fm* F25	*E.fm* F20	*E.fl* M19	*E.fl* M48	*E.fl* Se112
IS16	-	-	-	-	-	-	-	-	-	-	-
gelE	-	-	-	-	-	-	-	+	+	+	+
sprE	-	-	-	-	-	-	-	+	+	+	+
cylLL	-	-	-	-	-	-	-	+	+	+	+
cylLL/LS	-	-	-	-	-	-	-	-	-	+	+
cylM	-	-	-	-	-	-	-	-	-	-	-
cylB	-	-	-	-	-	-	-	-	-	-	-
hyl	-	-	-	-	-	-	-	-	-	-	-
esp	-	-	-	-	-	-	-	-	-	-	-
agg	-	-	-	-	-	-	-	-	+	+	-
asa1	-	-	-	-	-	-	-	-	+	+	-
ace	-	-	-	-	-	-	+	+		+	+
acm	-	-	-	-	-	-	+	+	+	+	+
EfaAfs	-	-	-	-	-	-	-	-	+	+	+
EfaAfm	+	+	+	+	+	+	+	+	-	-	-
kat	-	-	-	-	-	-	-	-	-	-	+
cpd	-	-	-	-	-	-	-	+	+	+	+
eep	-	-	-	-	-	-	-	+	+	+	+
ccf	-	-	-	-	-	-	-	-	-	-	-
cob	-	-	-	-	-	-	-	-	-	+	+
Lipfm	-	-	-	-	-	-	-	-	-	-	-
Lipfl	-	-	-	-	-	-	-	-	+	+	+
hdc		-	-	-	-	-	-	-	-	-	-
tdc		-	-	-	-	-	-	-	-	-	-
odc	-	-	-	-	-	-	-	-	-	-	-

[a]Control probiotic strain

probiotic safety [47]. The phenotypic results indicate that one strain (*E. faecium* Se83) shows resistance to ampicillin and erythromycin with a concentration of 16 and 15 µg/mL, respectively, while one strain (*E. faecium* Se142) shows intermediate resistance to erythromycin (Table S2). Genotypically, only gentamycin resistance gene (*aac6-aph2*) was found in all tested strains, followed by tetracycline (*tetL*, *tetM*) and penicillin resistance genes (*pbp5*) which were not detected in *E. faecium* NF and *E. faecium* TWCST1. All other screened genes related to vancomycin, erythromycin, chloramphenicol, and linezolid were not detected in any of the selected strains; only the fluoroquinolone gene (*parC*) was observed in two strains (Table 2).

Assessment of Enterocins (Antimicrobial Potential) and Evaluation of CRISPR-Cas Genes

Phenotypic results of the two tested methods (i.e., spot over-lawn and spot overlay) show that the selected strains have the ability to produce enterocins and thus have antimicrobial potential against the indicator strains. The activity was measured in terms of clear zone formation around the strains. Generally, it was observed that *E. faecium* strains produce greater zones than *E. faecalis* strains and thus have the ability to produce more enterocins. The highest activity was shown by strain *E. faecium* TWCST1, *E. faecium* M30, and *E. faecium* F25 against *M. luteus* (ATCC10242), *E. hirae* (Food), *L. monocytogenes* (ATCC13932), and *P. mirabilis* (UTI) (Table S3). The molecular assessment of genes responsible for enterocins such as *EntA*, *EntB*, *EntQ*, *EntL50*, and *EntP* was also checked in the selected strains. The highest detected enterocin genes *EntP* and *EntQ* were detected in all the tested strains (100%), followed by *EntB* (81.8%), *EntA* and *EntSE-K4* (72.7%), and *EntL50* and *Entef1097* (63.6%). Enterolysin A (*entL*) and *Entef1071* were absent in all tested strains. The highest number of enterocins genes was detected in strains *E. faecium* Se142 and *E.*

Table 2 Molecular assessment of antibiotic resistance genes in the selected enterococcal strains

Antibiotics	Genes	Selected enterococcal strains						
		E.fm NF[a]	*E.fm* TWCST1	*E.fm* TWCST2	*E.fm* M30	*E.fm* Se83	*E.fm* Se142	*E.fm* F25
Gentamycin (GEN)	*aph(3)IIIa*	+	+	+	+	+	+	+
	aac6-aph2	-	-	+	+	+	+	+
	VanC1/C2	-	-	-	-	-	-	-
Vancomycin (VAN)	*VanC3/C4*	-	-	-	-	-	-	-
	Van A	-	-	-	-	-	-	-
	VanB	-	-	-	-	-	-	-
Ampicillin (AMP)	*Pbp5*	-	-	+	+	+	+	+
Chloramphenicol (CH)	*Cat*	-	-	-	-	-	-	-
Erythromycin (ER)	*ermB*	-	-	-	-	-	-	-
	ant-(4)-Ia	-	-	-	-	-	-	-
Fluoroquinolone (FQ)	*gyrA*	-	-	-	-	-	-	-
	parC	-	-	-	-	+	+	-
Tetracycline (TET)	*TetL*	-	-	+	+	+	+	+
	TetM	-	-	+	+	+	+	+
Linezolid (LZ)	*Cfr*	-	-	-	-	-	-	-
	L3	-	-	-	-	-	-	-
	L42	-	-	-	-	-	-	-

[a]Probiotic control strain

faecium Se83 followed by *E. faecium* M30 and *E. faecium* TWCST2 strain (Table 3).

Molecular assessment of the selected strains confirms the presence of the CRISPR-Cas1 gene in all strains except *E. faecium* M30 and *E. faecium* F25, while CRISPR3 and CRISPR-Cas3 csn1 were detected in all strains except *E. faecium* NF (probiotic control). CRISPR2, which has variable PCR product size, was only detected in *E. faecium* M30 with an approximate size of 1400 bp. CRISPR-Cas1 csn1 was also detected in *E. faecium* TWCST2, *E. faecium* Se142, and *E. faecium* F25 (Table 4).

In Vitro Assessment for Probiotic Validation

The proposed probiotic strains were also subjected to various in vitro assays to assess their tolerance abilities, aggregation properties, and survivability. The ability of strains to tolerate such conditions in vitro can also tolerate in the animal host [48]. Similarly, adherence to intestinal mucosa and gut colonization were also checked by their adherence ability to HT-29 cells.

Table 3 Molecular assessment of enterocin-producing genes in the selected enterococcal strains

Strains	Enterocin genes								
	EntA	*EntB*	*EntP*	*EntQ*	*EntL50*	*Ent1097*	*Ent1071*	*Ent SE-K4*	*EntL*
E.fm NF[a]	+	+	+	+	-	-	-	-	-
E.fm TWCST1	+	+	+	+	+	-	-	-	-
E.fm TWCST2	+	+	+	+	+	+	-	-	-
E.fm M30	+	+	+	+	+	-	-	+	-
E.fm Se83	+	+	+	+	+	+	-	+	-
E.fm Se142	+	+	+	+	+	+	-	+	-
E.fm F25	-	+	+	+	+	-	-	+	-
E.fm F20	+	+	+	+	-	+	-	+	-
E.fl M19	-	+	+	+	+	+	-	+	-
E.fl M48	-	-	+	+	-	+	-	+	-
E.fl Se112	-	-	+	+	-	+	-	+	-

[a]Probiotic control strain

Table 4 Molecular genotyping of CRISPR-Cas-associated genes detected in the selected enterococcal strains

Genes	Strains					
	E.fm NF[a]	*E.fm* TWCST1	*E.fm* TWCST2	*E.fm* M30	*E.fm* Se142	*E.fm* F25
Crispr1	-	+	+	-	+	-
Crispr2	-	-	-	Vs	-	-
Crispr3	-	+	+	+	+	+
Crispr1 cas-csn1	-	-	+	-	+	+
Crispr3 cas-csn1	-	+	+	+	+	+

[a]Probiotic control strain

Vs variable PCR product size

Tolerance to Gastric Juice, Acids, and Salts

The gastric juice tolerance result shows that the selected strains were able to survive in the presence of gastric juice at low pH 1.0, 2.0, 3.0, and 4.0. All strains show the same pattern of tolerance except *E. faecium* TWCST1 and *E. faecium* M30 which show greater tolerance, while *E. faecium* TWCST2 shows low tolerance at pH 1.0 and 2.0 (Fig. 3).

Acid tolerance was checked against different pH (i.e., 2.0, 4.0, 7.0, 9.0, and 12.0). The results show maximum survivability and growth at pH 7.0 and 9.0 followed by pH 12.0. However, the bacterial growth was restricted at pH 2.0 and 4.0 (Fig. S4).

Treating against different concentrations of NaCl (1.0%, 3.0%, 5.0%, and 7.0%) shows that all strains have the ability to survive and tolerate high salty conditions, but the

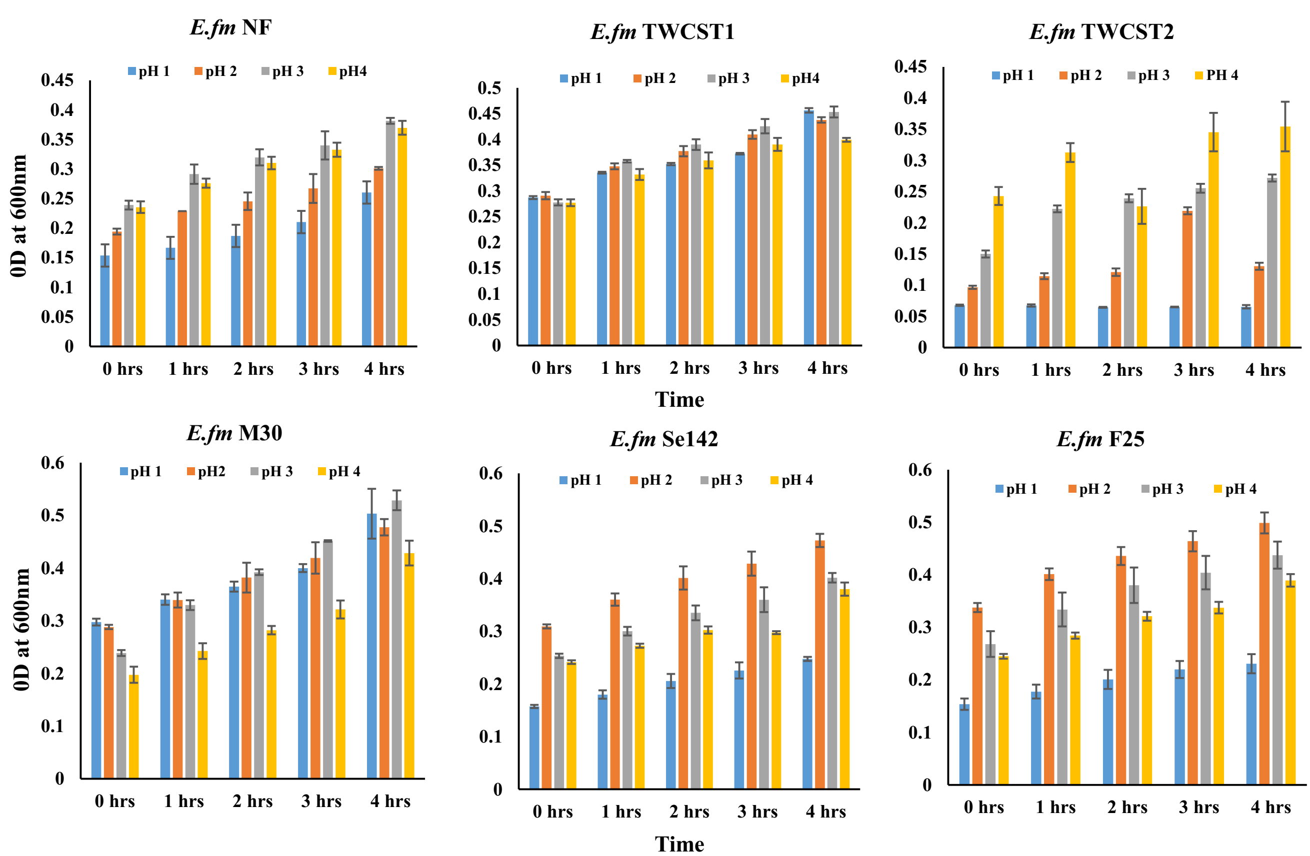


Fig. 3 Resistance of the selected enterococcal strains to the artificial gastric juice containing pepsin and acidified to pH 1.0 to pH 4.0 followed for 4-h incubation at 37 °C (mimicking transient time of food in the stomach)

growth starts to decrease at 7.0% NaCl, suggesting that our strain can significantly tolerate NaCl up to 5% concentration (Fig. S5).

Auto- and Co-aggregation and Cell Surface Hydrophobicity

Co-aggregation results reveal that the selected strains show variable percentage against different indicator strains and range from – 36 to 46%. After 24 h of incubation, high co-aggregation was observed between *E. faecium* NF and *L. monocytogenes* and *E. faecium* F25 with *Proteus mirabilis* (46%), followed by *E. faecium* TWCST1 and *E. faecium* M30 with *Lactobacillus* spp. (39%), as shown in the form of heat map (Fig. 4).

Similarly, selected strains show high-level (˃ 50%) auto-aggregation as all having value greater than 75% after 1 h of incubation. The highest value was shown by *E. faecium* TWCST1 and *E. faecium* Se142 (82%) followed by *E. faecium* NF (80%). The lowest auto-aggregation was shown by *E. faecium* F25 (77%) which is isolated from food source (Fig. 5A).

Hydrophobicity of the selected strains shows the binding capacity from 18 to 69% with xylene. The highest value was observed for *E. faecium* NF (69%) and *E. faecium* F25 (63%) followed by *E. faecium* TWCST1 (59%), while the lowest value (18%) was observed in *E. faecium* TWCST2 (Fig. 5B).

Cytotoxicity and Adherence Assay

Cytotoxicity was not observed in any of the selected strains, which was also confirmed by the absence of hemolytic activity. The qualitative adherence against HT-29 cells indicates that *E. faecium* M30 has the ability to adhere with cells as demonstrated in Fig. 6.

Discussion

Probiotics are live microorganism which can help in the proper functioning of our body system like improve immunity, kill pathogens, enhance gut colonization, and decrease cholesterol level [19, 33, 49]. To select potential strains to be used as probiotics, various guideline are proposed by the EFSA, FAO, and WHO, including proper identification, absence of antibiotic resistance and virulence genes, tolerance and colonization ability, aggregation properties, and production of antimicrobial substances [12, 17]. Like other LAB probiotics, enterococci are also used with careful assessment [17, 23]. Enterococcal probiotics have advantages for human usage as these are the natural resident of human GIT. *E. faecium* SF-68 is a pretty known example and available commercially for the treatment of various disease. Other enterococcal probiotics are used in poultry, swine, domestic, and cattle farming industry [6]. The ability of harboring virulence genes and antibiotic resistance transferable genes makes them debatable for their probiotic potential [3, 6, 50]. But as common nosocomial microbes, having genome plasticity and the ability to withstand various harsh conditions and their role in various industries make them a suitable candidate for this potential [24, 34].

To identify and validate potential probiotic strains from the locally collected enterococcal strains, one hundred and twenty-two strains were selected and subjected for their initial screening. Various virulence traits like gelatinase, lipase, urease, catalase, DNAse, and hemolysin production were carried out and critically analyzed and found negative [21]. All these potential virulence traits are involved in different diseases and can enhance the pathogenicity of enterococci. For instance, gelatinase, which is an endopeptidase enzyme, can cleave gelatin and associated proteins [33, 51], biofilm formation aid to the pathogenicity [49], and hemolysin production which is the most dangerous virulence trait and

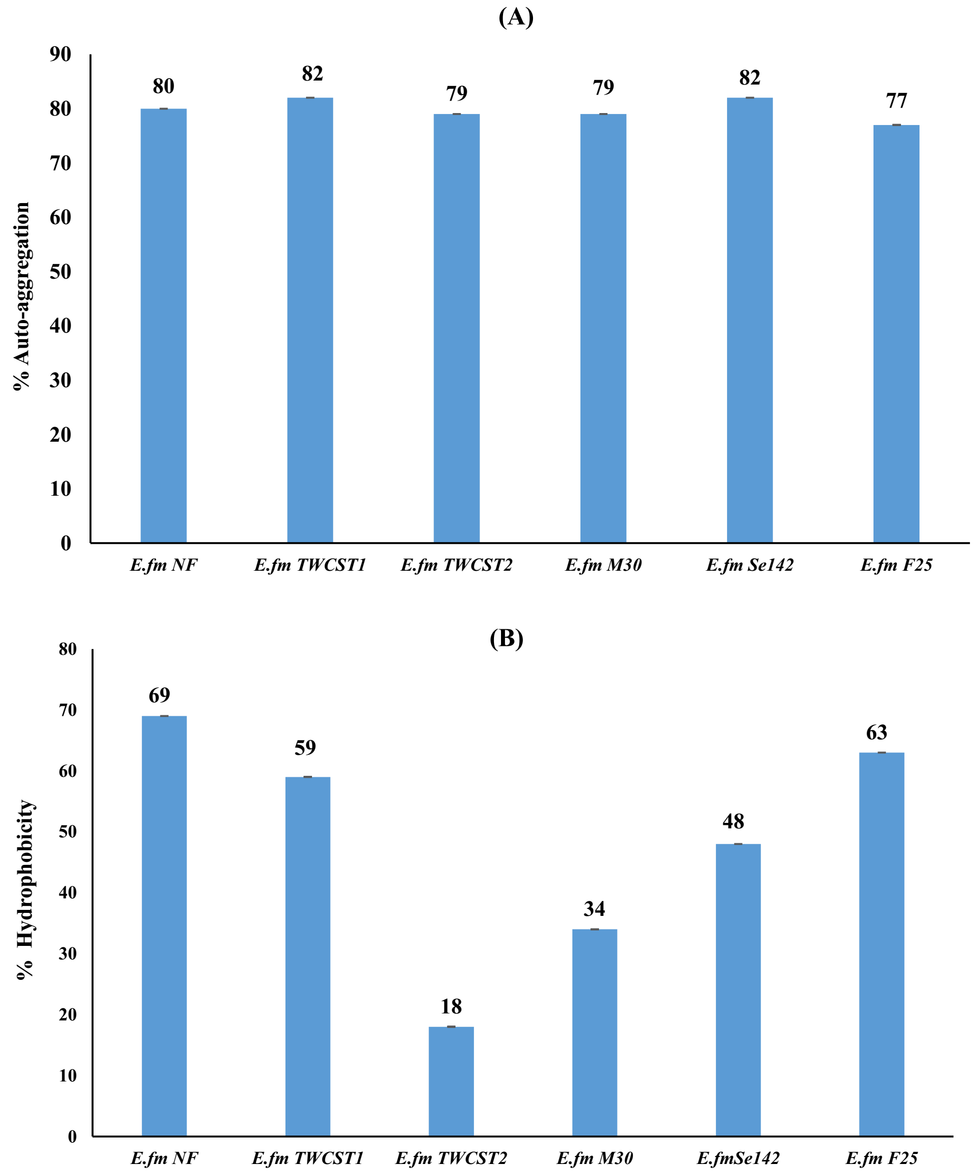
Fig. 4 Heat map constructed for percent co-aggregation of the selected enterococcal strains with *L. monocytogenes*, *Proteus mirabilis*, *Lactobacillus*, *Sensitive*, and *Micrococcus luteus strains*

	Co-aggregate partner					
strains	LM	PM	LB	SEN	ML	E.coil
E.fm NF	46	7	18	21	7	-1
E.fm TWCST1	-1	4	39	-5	-22	9
E.fm TWCST2	-22	-12	32	24	5	12
E.fm M30	-14	3	38	29	25	10
E.fm Se142	0	8	-9	5	-5	-2
E.fm F25	0	46	27	-36	21	4

Scale: 46, 39, 38, 32, 29, 27, 25, 24, 21, 18, 12, 10, 9, 8, 7, 5, 4, 3, 0, -1, -2, -5, -9, -12, -14, -22, -36

LM; Listeria monocytogenes, PM; Proteus mirabilis, LB; lactobacillus, SEN; enterococcal sensitive, ML; Micrococcus luteus

Fig. 5 Figure illustrating the percent auto-aggregation and cell surface hydrophobicity of the selected enterococcal strains. **A** Auto-aggregation; **B** hydrophobicity. Cell surface hydrophobicity was performed against xylene (hydrocarbon)



involved in endocarditis and lysis of the eukaryotic red blood cells [6, 33]. It is also reported that if a strain is free from hemolysin production, then it may be used as probiotic [47]. In this study, human blood was used as it is more sensitive than other blood sources [49]. The absence of cytotoxicity not only confirms the absence of hemolysin production but also makes the strains safer. Based on aforementioned result, only eleven strains ($n=11$) were selected that were negative for all potential virulence traits and followed further for their molecular assessment.

It is worthy to mention that for probiotic evaluation, the phenotypic absence of any virulence traits is not enough until their absence was confirmed at molecular level. The molecular assessment of different genes like insertion sequence (*IS*16) which is considered a key identifier agent in multidrug resistance [11], aggregation substance genes which are present in plasmid and involved in colonization and have the chance of transferring to the host [31], *esp* gene involved in biofilm formation, and hyl gene responsible for pathogenic infection and sex pheromones genes [50] were screened in the selected strains and not found in them. Genes responsible for adhesion, i.e., *EfaAfm* and *EfaAfs* in *E. faecium* and *E. faecalis*, respectively, were detected in their respective species. Similar results for virulence genotyping of enterococcal probiotics were also shown in the results of [264352. Biogenic amines are low molecular weight organic compound and considered an unwanted property for probiotics [53], were also absent in the selected strains, and hence make them more suitable for their use in food science. Irrespective of the phenotypic absence of virulence traits, there were some virulence

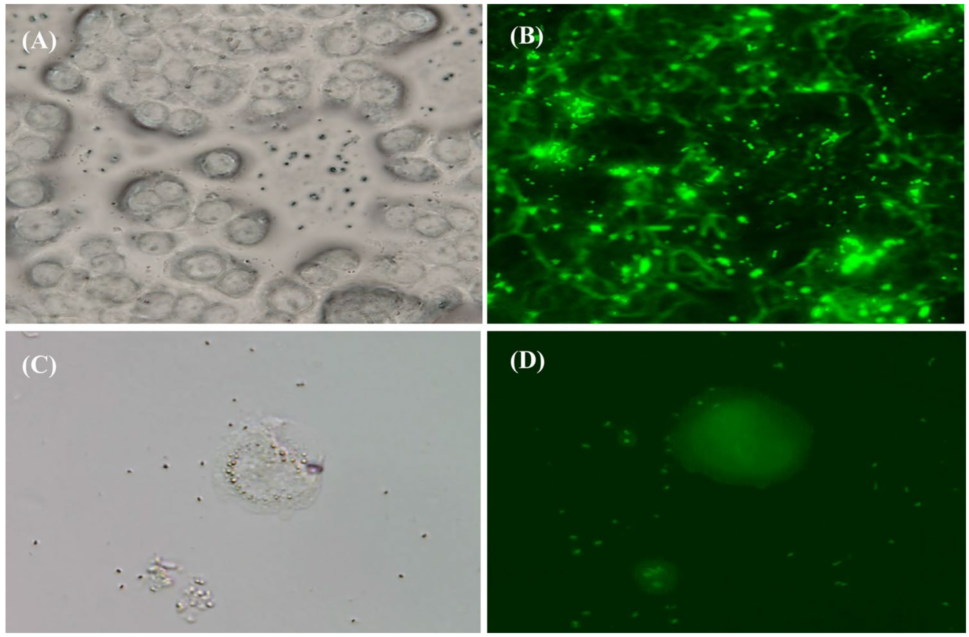


Fig. 6 Microscopic images showing adherence of *E.fm* M30 strain with HT-29 cells line. **A** Bright-field adherence with cells line in multiple. **B** FITC-labeled cell adherence. **C** Bright-field adherence around single cell. **D** FITC-labeled single cell adherence

genes identified in the *E. faecalis* strains, so these affected strains were discarded from further analysis.

Antibiotic resistance evaluation is of utmost importance in the case of enterococci because some strains harbor resistance transferring genes and have the ability to transfer them into other pathogenic bacteria in the gut [33]. Enterococci have both intrinsic and acquired antibiotic resistance genes which make them suspected, and for probiotic use, these genes must be critically analyzed [49]. According to Suvorov et al. [26], the intrinsic resistance genes are present on chromosome and cannot be transferred easily so the presence of such genes does not affect the probiotic potential. Molecular assessment of antibiotic resistance genes confirms the presence of gentamycin and tetracycline resistance genes; these genes were also observed in the study [26, 54]. All other antibiotic resistance genes, likes genes of vancomycin which is the most important in probiotic safety, erythromycin, chloramphenicol, and linezolid, were completely absent in the tested strains.

The genetic heterogeneity by RAPD-PCR [43], protein fingerprinting via SDS-PAGE [39], and probiotic-specific peak by ATR-FTIR [46, 55] was also identified in the strains. Next-generation probiotic development strategies are used to enhance animal growth and feed formulation optimization/conversion and for the competitive exclusion of pathogenic bacteria [56]. CRISPR-Cas genes are found in the plasmid of *Lactobacilli* and can help them to promote their genetic stability. Recently, it is also reported that there is an inverse relationship between antibiotic resistance gene, cytolysin, *cob*, and aggregation substance, with CRISPR-Cas genes [57]. CRISPR-Cas also have a role in the acquisition prevention of pathogenicity island (PI), thus helping the strains to reject any virulence traits [58]. In this study, CRISPR-Cas genes were detected with variable frequency thus making the strain more suitable for their probiotic potential.

Antimicrobial activities are considered the most important character for probiotic strains and were also evaluated both phenotypically in terms of formation of zone of inhibition against indicator strains and at molecular level in terms of enterocin-producing genes. The phenotypic results show that all tested strains possess antimicrobial activity against indicator strains with different zones of inhibition. The highest zone of inhibition was detected against *E. hirae* followed by *L. monocytogenes*. The detection of multiple enterocin genes in the selected strains can increase the chances of their probiotic potential as this character is mainly because of enterocins [32]. *EntP* and *EntQ* were the most detectable enterocins followed by *EntA*, *EntB*, and *EntL50*. The specific antimicrobial activity against *Listeria* mostly occurs due the *EntA* production. It is important to note that Entef1097 and EntSE-K4 were mostly observed in *E. faecalis* strains while EntA and EntL50 were mostly detected in *E. faecium* strains.

Various tolerance assays and aggregation properties were checked in vitro by different assays, like acid, NaCl, and gastric juice tolerance; aggregation properties like auto- and co-aggregation; and cell surface hydrophobicity. It is pretty known that if the bacteria can tolerate such harsh in vitro conditions, then they can also survive in the GIT [56, 59]. In tolerance assays, all tested strains show good tolerance activity against provided conditions; therefore, all these strains have the probiotic potential. The intestinal mucosa is the most exposed part to the gut microorganism and any imbalance can cause severe complications [48]. Therefore, adhesion of probiotics to mucosal surface is one of the top criteria in the selection of probiotics [60]. Adhesion is related to both auto-aggregation and hydrophobic properties [25]. Aggregation properties of probiotic strains can help in colonization and adherence and help in bacterial equilibrium of the gut [33, 59]. According to Baccouri et al. [49], hydrophobicity and auto-aggregation help in adhesion to intestinal cell and prevent pathogen colonization; thus, these are considered beneficial properties of probiotics.

Auto-aggregation is the self-clumping of bacteria which can help them to increase their mass and eradicate the pathogens [33]. In this regard, these strains show high-level auto-aggregation (˃ 50%) ability as high as 82%, which makes them suitable probiotic candidates. Similarly, co-aggregation is another key feature of probiotics which helps them to kill pathogenic bacteria in the proximity of probiotic strains by reducing cell–cell distance [6] and also create a barrier for pathogens to bind with epithelial cells [33]. Fugaban et al. [6] recently reported that higher co-aggregation (˃ 50%) is lethal when the co-aggregate partner is an enterocin-resistant pathogen. Another potential probiotic feature is the cell surface hydrophobicity which helps the bacteria to bind to epithelial cells and helps in colonization. Higher hydrophobicity reflects the potential of probiotic strain [40]. The tested strains show higher values of hydrophobicity which makes them potential probiotic candidates.

Adherence ability of probiotic microorganism is also considered major selection criteria which helps them in their colonization in the gut [49]. The adhesion property is strain specific as different strains show different molecules for their adherence [44]. The results reflect that the selected strains have the ability to adhere with HT-29 cell line. The harmless nature of bacteria is necessary criteria for probiotics [44] which is checked by cytotoxicity assay. There was no cytotoxicity observed in the tested strains and hence makes them promising lead probiotic strains.

Conclusion

The ubiquitous distribution, genomic plasticity, and production of antimicrobial substances make the genus *Enterococcus* as a potential microorganism in different applications. Being not included in GRAS and QPS status, very careful evaluation must be taken to check their pathogenic nature. Although there are some commercially available enterococcal probiotics that are used for the treatment of various diseases, in this study, various phenotypic and genotypic assessments of virulence traits, antibiotic resistance, and in vitro conditions that mimic the host gut environments were carried out to check the safety and tolerance of the tested strains. Based on these aforementioned conditions, it is concluded that all the selected strains have the potential that may be used as probiotics or as feed additive for animals.

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Author Contribution Conception, design, and supervision were performed by SAA. Material preparation, data collection, and analysis were performed by AHn, SA, DA, MR, AA, and SAA.

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Data Availability All data generated or analyzed during this study are included in this published article and the supplementary information available on-line at the publisher website.

Declarations

Competing Interests The authors declare no competing interests.

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